An example showcasing the features of the msheet package

1 The msheet class (msheet.cls)

The icon on the left hand side of the title box is set according to the chosen document type. Available options are *worksheet*, *factsheet*, *groupwork*, *schedulework* and *experiment*. This document is of type *factsheet*.

We can force a rounded title box using the option *rounded*. The background color of the title box might be chosen as well. Available options are *cyan*, *grey*, *white* and *yellow*.

The line spacing can be set via the options *singlespacing*, *onehalfspacing* (this is the setting in this document) and *doublespacing*.

All of these can also be set with `\msheetsetup`, which works in the preamble and in the document body: `\msheetsetup{type=experiment, titlecolor=DarkCyan}`. The old public `\icon` and `\titlebackcolor` commands no longer exist.

A license and an attribution might be given via `\license{}` and `\attribution{}` respectively (see footer). Both are optional; if they are omitted, the corresponding part of the footer is simply left out and a single warning is issued.

2 The msheet style (msheet.sty)

2.1 Headings

`\mchapter` and `\mtestpart` typeset centred coloured headings:

Ein Kapitel

Ein Teil

2.2 Exercise headings

Aufgabe 1 *A subsection like task environment.*

Compute the average velocity between $t = 5$ and $t = 8$.

1. $f(t) = 6t$ **2.** $f(t) = 6t + 2$ **3.** $f(t) = \frac{1}{2}at^2$

4. $f(t) = t - t^2$ **5.** $f(t) = 6$

Aufgabe *And the same as starred version.*

Compute the average velocity between $t = 5$ and $t = 8$.

a) $f(t)=6t$ **b)** $f(t)=6t+2$ **c)** $f(t)=\frac{1}{2}at^2$

d) $f(t)=t-t^2$ **e)** $f(t)=6$

Aufgabe 2 *A paragraph like task environment.*

a) $f(t)=6t$ **b)** $f(t)=6t+2$ **c)** $f(t)=\frac{1}{2}at^2$

d) $f(t)=t-t^2$ **e)** $f(t)=6$

Aufgabe *And the same as starred version.*

1. $f(t)=6t$
2. $f(t)=6t+2$
3. $f(t)=\frac{1}{2}at^2$
4. $f(t)=t-t^2$
5. $f(t)=6$

The old names `\msexercise` and `\mpexercise` still work; they are aliases of `\msection` and `\msectionr`.

»» Bitte wenden! »»

2.3 Lists

1. When you jump and fall back your height is $y = 2t - t^2$ in the right units.

1.1. Graph this parabola and its slope.

1.2. Find the time in the air and the maximum height.

1.3. Prove: Half of the time you are above $y = \frac{3}{4}$. tip 1 solution 1

Basketball players hang in the air partly because of **1.3.**.

2. If the ball on the unit circle reaches t degrees at the time t, find its position and speed and upward velocity. recommended

Aufgabe 1 When you jump and fall back your height is $y = 2t - t^2$ in the right units.

a) Graph this parabola and its slope.

i. Graph this parabola and its slope.

b) Find the time in the air and the maximum height.

c) Prove: Half of the time you are above $y = \frac{3}{4}$. tip 1 solution 1

Basketball players hang in the air partly because of **c)**.

Aufgabe 2 If the ball on the unit circle reaches t degrees at the time t, find its position and speed and upward velocity. recommended

A plain enumeration with the msheet labels:

- **a)** first
- **b)** second
  - (1) nested

Section-numbered enumeration (`menumeratex`):

- 2.1. first
- 2.2. second
  - 2.2.1. nested

And the ticked list:

👍 Hausaufgabe abgegeben 👍 Heft nachgeführt

»» Bitte wenden! »»

2.4 Boxes

Theorem 1 – l'Hôpitals rule

Suppose $f(x)$ and $g(x)$ both approach zero as $x \to a$. Then $\frac{f(x)}{g(x)}$ approaches the same limit as $\frac{f'(x)}{g'(x)}$, if that second limit exists:

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}.$$

Experiment – The gravitational force

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Example

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

A golden titled box

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

An untitled red box.

An untitled green box.

An untitled blue box.

An untitled box in an arbitrary `svgnames` colour.

2.5 Inline boxes

The inline boxes are meant for hints and markers inside running text: Achtung , Tipp , Beispiel and beliebige Farbe . `\mibnp` puts a „Bitte wenden!“ marker at the foot of the page and breaks it.

2.6 Scores

`\mscore` puts the score of an exercise into the margin and adds it to a running total; `\mtotalscore` prints that total. `\mresetscore` starts a new count.

Aufgabe 3 *Scored exercise*

Compute $\int_0^1 x^2 \, dx$. **4 BE**

Aufgabe 4 *Another scored exercise*

Compute $\int_0^1 x^3 \, dx$. **2.5 BE**

$\overline{\textbf{6.5 BE}}$

`\mresetscore` starts a new count, for a sheet with several independently scored parts.

Aufgabe 5 *A third exercise, counted separately*

Compute $\int_0^1 x^4 \,\mathrm{d}x$. **3 BE**

$\overline{\textbf{3 BE}}$

Viel Erfolg!